

Jared King

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🌐 [linkedin.com/JaredKing](https://www.linkedin.com/JaredKing)

🐙 github.com/JaredKing

🌐 [Portfolio/website](#)

Technical Skills

Languages: Python, C, C++, Swift, SQL, JavaScript, TypeScript

Frameworks: Pandas, Flask, React.js, Dash, Django, NumPy, SwiftUI, AWS Amplify, AWS CLI

Technologies: AWS (S3, EC2, Lambda, Cognito, IAM), Docker, Jupyter, Terraform, PostgreSQL, Snowflake, Git, Linux

Experience

Software Developer - Trading Systems

Sept 2024 – Dec 2024

Castleton Commodities International

Calgary, Alberta

- Constructed library with Pandas, SQLAlchemy, and s3fs to optimize Snowflake and AWS S3 connections for multi-threaded quantitative application, ensuring thread safety and reducing query times by 50%.
- Created development pipeline using Docker, EC2, Terraform, and Bitbucket Pipelines to allow for isolated deployment tests.
- Improved Docker builds by optimizing C library compilations, cutting build time by 75%, saving 40-60 hours monthly.
- Led Python 3.8 to 3.11 upgrade for mission critical models, using test-driven development to ensure system stability
- Optimized Pandas-based trading algorithms with query caching and overhead reduction, cutting execution time from 8 to 3 hours and memory usage by 40%, while monitoring performance using Datadog and AWS CloudWatch.

Software Developer - GIS Applications

Jan 2024 – Apr 2024

The Solution Stack

Calgary, Alberta

- Developed iOS Application with Swift, AWS Amplify, and ArcGIS, enabling users to receive real-time alerts and visualizations of active fire hazards in Western Canada through map interface.
- Crafted queue in Swift, leveraging native iOS file system and network monitor to manage and synchronize events through a RESTful Lambda API, enabling 100% app functionality while offline.
- Engineered web application for user management using React.js, TypeScript, AWS Amplify gen 2, and the AWS SDK, reducing user management time across organizations by 80%.
- Instantiated CI/CD pipeline for Event Bridge AWS Lambdas, reducing deployment time by 85%.
- Utilized feedback in an Agile workflow to deploy into production with 100% uptime for over 100 users.

Software Developer - Aviation Management Systems

May 2023 – Aug 2023

Unity Aviation Canada

Airdrie, Alberta

- Spearheaded company-wide productivity tool from concept to implementation using Microsoft SQL, Django, HTML, CSS, and JS, centralizing important company documents while improving requisition times by 90%.
- Migrated 15+ years of company data into a new SaaS system using T-SQL, ensuring data integrity and availability.
- Implemented unit tests, authentication and authorization, and segregated environments to ensure system reliability before deploying locally with Apache and WSGI.

Extracurriculars

Software Developer - Motor Controls

Apr 2024 – Sept 2024

Electrium Mobility Design Team

Waterloo, Ontario

- Implemented real time browser-controlled RC car utilizing the Web Socket protocol on a ESP8266 micro-controller that resulted in 3rd/ 20 in RC competition.
- Developed PWM module with Arduino in C++, enabling 10-stage control over electric motor.

Projects

🔗 OfficeBids | *React.js, Bootstrap, Supabase, Vercel, PostgreSQL*

- Created web app utilizing Supabase, PostgreSQL, JavaScript, React.js, and Bootstrap enabling employees to vote/bid on new office items during office space relocation.
- Implemented CI/CD pipeline with AWS Codecommit and Vercel to deploy the application for a small companies usage.

🔗 High Frequency Trading Comparison | *Python, NLTK, Alpha Vantage, FinBERT*

- Developed a multi-threaded HFT system using Alpha Vantage API, integrating VADER and FinBERT, and a LSTM model for real-time trading insights with 60% accuracy for stock prediction.

🔗 Multi-process Web-Crawler | *C, CURL, OOP, Data Structures and Algorithms, IPC, Ubuntu*

- Designed and implemented a multi-process web crawler in C utilizing a hash table and stack to efficiently processes data, solving the producer-consumer problem using inter-process communication.

Education

University of Waterloo

Sept 2022 - Current

Candidate for Computer Engineering, Honours, Co-operative Program. Term 3A

Waterloo, Ontario

- **Relevant Coursework:** DSA (C++), Algorithms (Python), Operating Systems (C), System Programming and Concurrency (C)